Example 9.1

Calculate Karl Pearson's coefficient of correlation from the following data:

X:	6	8	12	15	18	20	24	28	31
Y:	10	12	15	15	18	25	22	26	28

Solution:

X	$x = (X-18)$	x^2
6	–12	144
8	–10	100
12	–6	36
15	–3	9
18	0	0
20	2	4
24	6	36
28	10	100
31	13	169
$\sum X = 162$	$\sum x = 0$	$\sum x^2 = 598$

Y	$y = (Y-19)$	y^2	xy
10	–9	81	108
12	–7	49	70
15	–4	16	24
15	–4	16	12
18	–1	1	0
25	6	36	12
22	3	9	18
26	7	49	70
28	9	81	117
$\sum Y = 171$	$\sum y = 0$	$\sum y^2 = 338$	$\sum xy = 431$

Table 9.1

$N=9,\ \overline{X} = \frac{\Sigma X}{N} = \frac{162}{9} = 18, \overline{Y} = \frac{\Sigma Y}{N} = \frac{171}{9} = 19$

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}}$$

where $x=(X-\overline{X})$ and $y=(Y-\overline{Y})$

$\Sigma xy = 431,\ \Sigma x^2 = 598,\ \Sigma y^2 = 338$

$$r = \frac{431}{\sqrt{598 \times 338}} = \frac{431}{449.582} = +0.959$$

(ii) When actual values are taken (without deviation)

when the values of X and Y are considerably small in magnitude the following formula can be used

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2} \times \sqrt{N\Sigma Y^2 - (\Sigma Y)^2}}$$

Example 9.2

Calculate coefficient of correlation from the following data

X	12	9	8	10	11	13	7
Y	14	8	6	9	11	12	3

Solution:

In both the series items are in small number. Therefore correlation coefficient can also be calculated without taking deviations from actual means or assumed mean.

X	Y	X^2	Y^2	XY
12	14	144	196	168
9	8	81	64	72
8	6	64	36	48
10	9	100	81	90
11	11	121	121	121
13	12	169	144	156
7	3	49	9	21
$\sum X = 70$	$\sum Y = 63$	$\sum X^2 = 728$	$\sum Y^2 = 651$	$\sum XY = 676$

Table 9.2

$$r = \frac{N\Sigma XY - (\Sigma X)(\Sigma Y)}{\sqrt{N\Sigma X^2 - (\Sigma X)^2} \times \sqrt{N\Sigma Y^2 - (\Sigma Y)^2}}$$

$$= \frac{7(676) - (70)(63)}{\sqrt{7(728) - (70)^2} \times \sqrt{7(651) - (63)^2}}$$

$$= \frac{322}{339.48}$$

$r = +0.95$

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